

Intro to ML

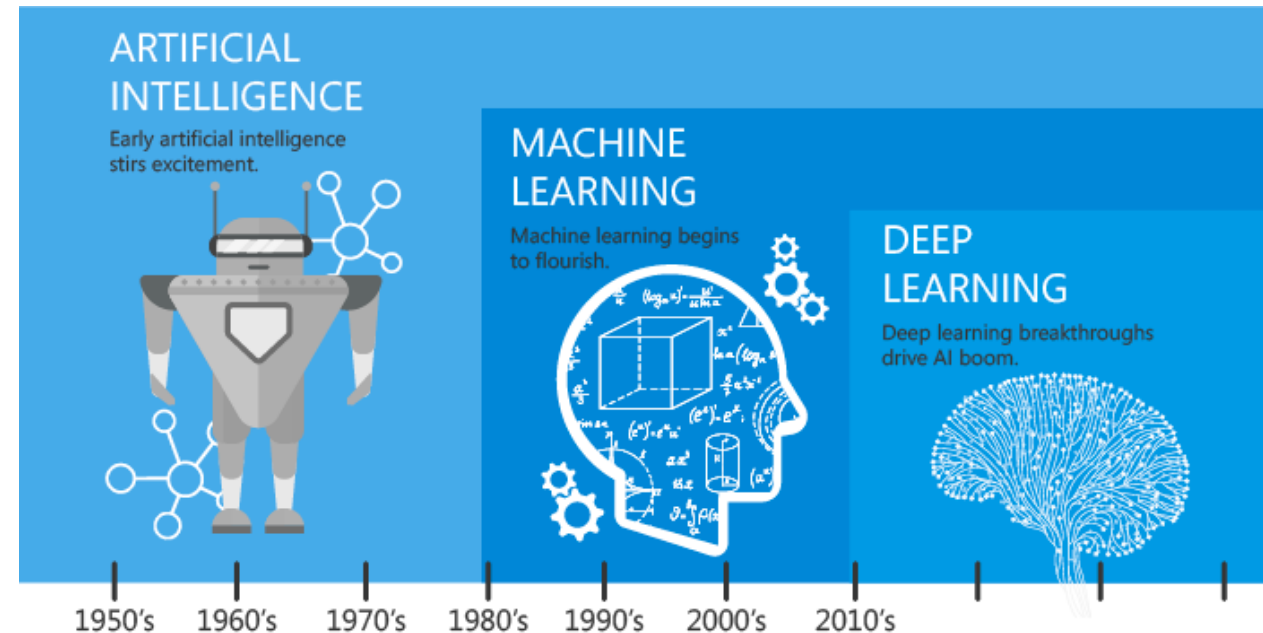
Gunhee Kim



서울대학교
SEOUL NATIONAL UNIVERSITY

Machine Learning (ML)

- A branch of **artificial intelligence**, concerned with the design and development of algorithms that allow computers to evolve behaviors based on empirical data
 - A ML algorithm is an algorithm that can ***learn*** from data
 - AI does not always imply a learning-based system
 - Symbolic reasoning
 - Rule based system
 - Tree search
 - etc.



What is Intelligence?

- Is there a “holistic” definition for intelligence?
 - The definition of intelligence is controversial
 - The ability to acquire and apply knowledge and skills
- We might list elements of intelligence:
 - Understanding, reasoning, problem solving, learning, common sense, generalizing, inference, analogy, recall, intuition, emotion, self-awareness
 - Which of these are necessary for intelligence? Which are sufficient?

What is AI?

- Views of AI fall into four categories:

Thinking humanly	Thinking rationally
Acting humanly	Acting rationally

- How does *humanly* differ from *rationally*?
 - Humanly:
 - (1) Solve the problems the same way humans do
 - (2) Require cognitive modeling
 - (3) Common-sense reasoning, social behavior
 - Rationally:
 - (1) Achieve goals by maximizing performance measure
 - (2) Use logic and deal with uncertainty and complexity

What is AI?

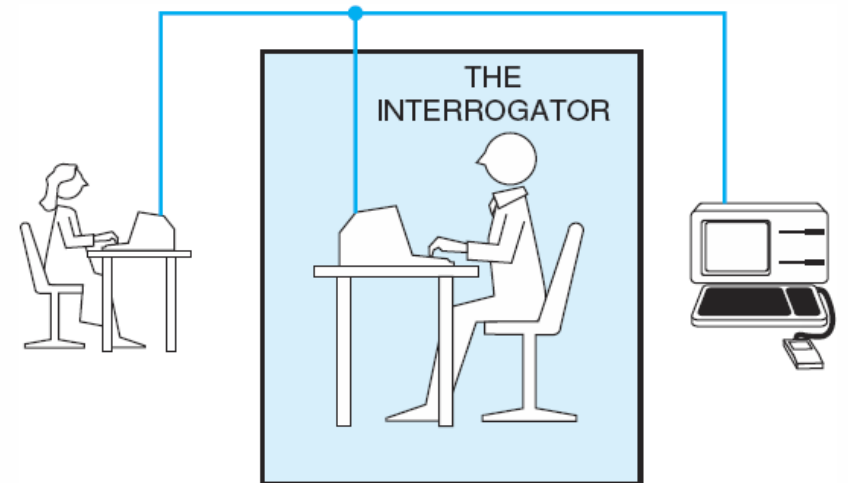
- Views of AI fall into four categories:

Thinking humanly	Thinking rationally
Acting humanly	Acting rationally

- How does *intelligent thinking* differ from *intelligent behavior*?
 - Intelligence is hard to define, but intelligence behavior is not
 - A machine has to display intelligent behavior that is indistinguishable from that of a human
 - Turing test!

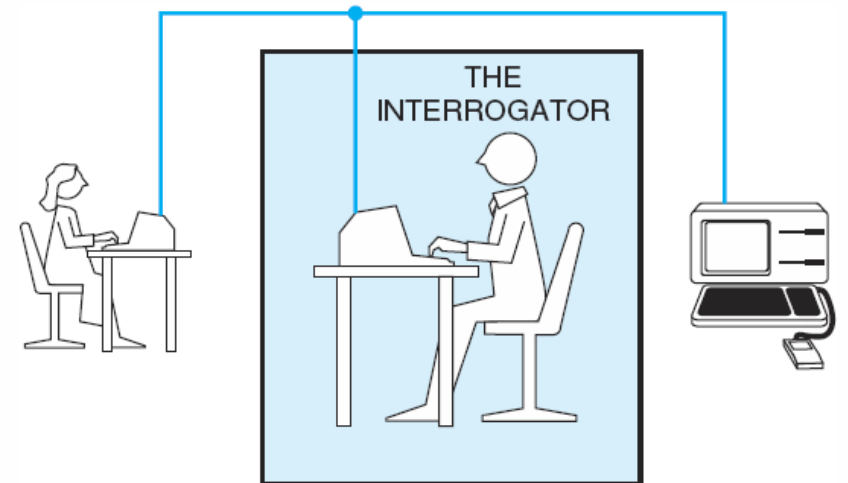
The Turing Test (1950)

- Alan Turing devised a test for defining AI
 - An interrogator poses questions to a human and a computer
 - If the interrogator cannot tell which is the human/computer, then the computer passes the Turing Test and should be considered intelligent
 - Turing first called this the *Imitation game* but has been renamed the Turing Test
- Can machines think?
 - Q: Please write me a sonnet on the subject of the Forth Bridge.
 - A: Count me out on this one. I never could write poetry.
 - Q: Add 34957 to 70764.
 - A: (Pause about 30 sec. and answer) 105621.



The Turing Test (1950)

- Acting humanly... NOT acting rationally
 - Tests behavior: simple and objective
 - Passing the Turing Test does not necessarily imply intelligence
 - Suggested major components of AI:
 - Knowledge representation
 - Automated reasoning
 - Learning
 - Language/image understanding (+ Robotics)



Strong AI vs Weak AI

- Weak AI, as opposed to strong AI, is also known as *applied* AI or *narrow* AI
 - Actions, decision, and ideas are programmed into it
 - All current forms of AI are weak AI
 - The weak AI hypothesis: the philosophical position that machines can demonstrate intelligence, but do not necessarily have a mind, mental states or consciousness

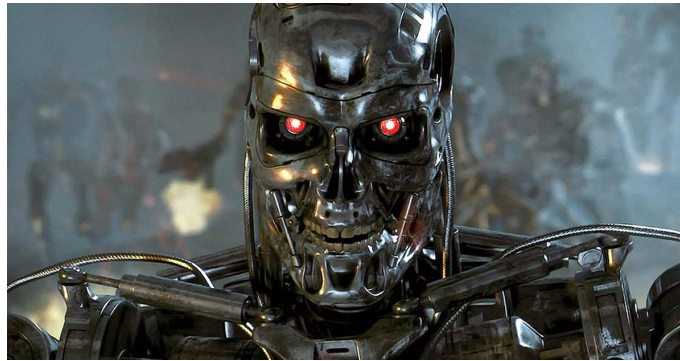


Strong AI vs Weak AI

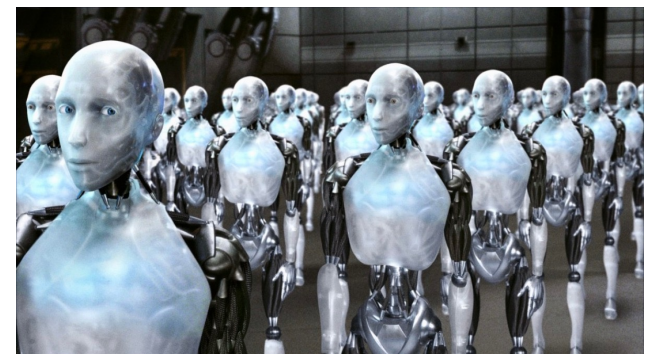
- Strong AI matches or exceeds human intelligence
 - Don't just simulate humans, they are intelligent on their own
 - Also referred to as *General AI*
 - Associates with such human traits as consciousness, sentience, sapience, free-will and self-awareness
 - e.g,. robots from the movies (HAL9000, Matrix, Terminator, I robot, etc)



Space Odyssey HAL 9000



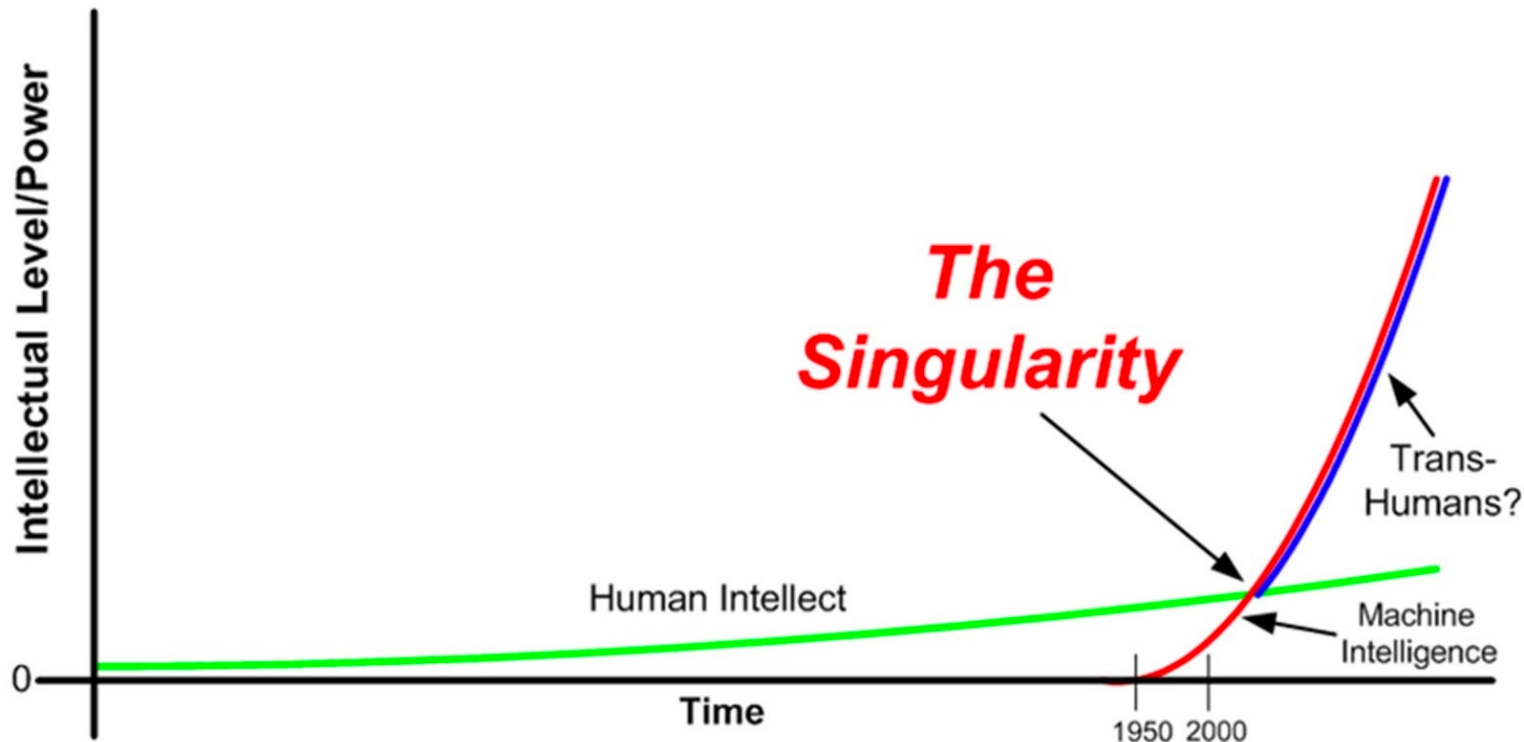
Terminator series



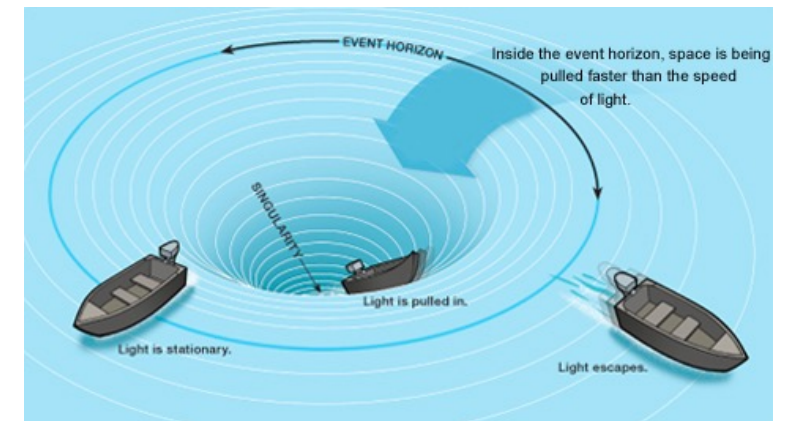
I, Robot

Technological Singularity

- Emergence of superhuman intelligence
 - Key idea: self-improvement
 - Some predict: this century, but others argue: never



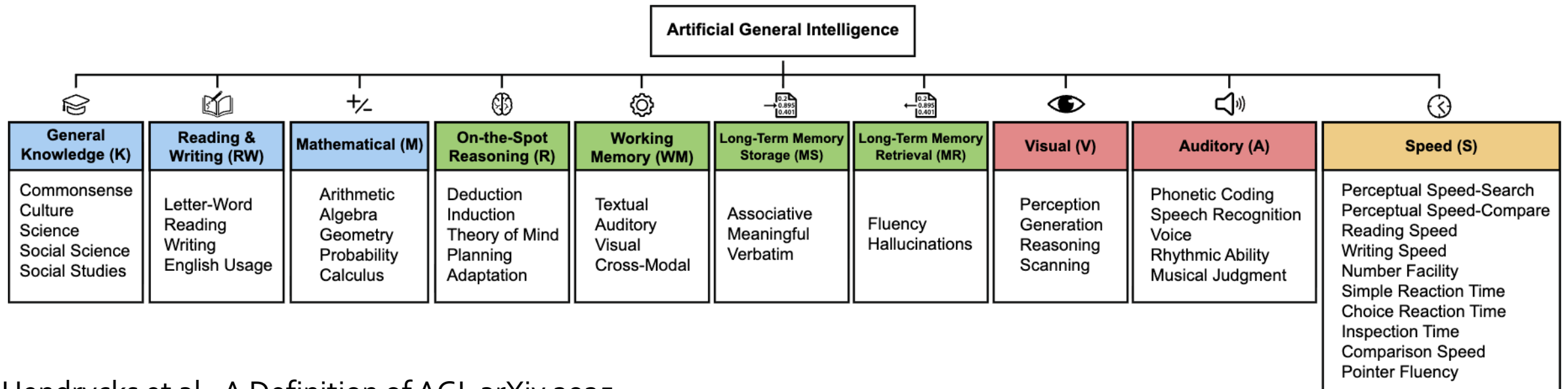
Analogy between *inability to predict* events after the development of a super-intelligence, and the space-time singularity beyond the event horizon of a black hole



What is AGI?

AGI is an AI that can match or exceed the cognitive versatility and proficiency of a well-educated adult.

- Ten core cognitive components of our AGI definition
 - Ground in the Cattell-Horn-Carroll (CHC) theory of cognitive abilities



What is AGI?

- Deepmind's view
 - 6 levels of AGI performance (depth), generality (width), and autonomy

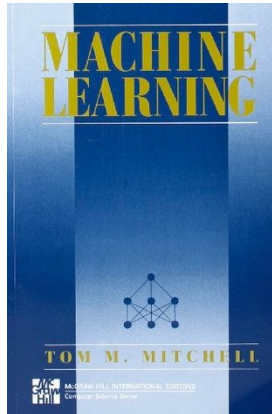
Performance (rows) x Generality (columns)	Narrow <i>clearly scoped task or set of tasks</i>	General <i>wide range of non-physical tasks, including metacognitive tasks like learning new skills</i>
Level 0: No AI	Narrow Non-AI calculator software; compiler	General Non-AI human-in-the-loop computing, e.g., Amazon Mechanical Turk
Level 1: Emerging <i>equal to or somewhat better than an unskilled human</i>	Emerging Narrow AI GOF AI (Boden, 2014); simple rule-based systems, e.g., SHRDLU (Winograd, 1971)	Emerging AGI ChatGPT (OpenAI, 2023), Bard (Anil et al., 2023), Llama 2 (Touvron et al., 2023), Gemini (Pichai & Hassabis, 2023)
Level 2: Competent <i>at least 50th percentile of skilled adults</i>	Competent Narrow AI toxicity detectors such as Jigsaw (Das et al., 2022); Smart Speakers such as Siri (Apple), Alexa (Amazon), or Google Assistant (Google); VQA systems such as PaLI (Chen et al., 2023); Watson (IBM); SOTA LLMs for a subset of tasks (e.g., short essay writing, simple coding)	Competent AGI not yet achieved
Level 3: Expert <i>at least 90th percentile of skilled adults</i>	Expert Narrow AI spelling & grammar checkers such as Grammarly (Grammarly, 2023); generative image models such as Imagen (Saharia et al., 2022) or Dall-E 2 (Ramesh et al., 2022)	Expert AGI not yet achieved
Level 4: Exceptional <i>at least 99th percentile of skilled adults</i>	Exceptional Narrow AI Deep Blue (Campbell et al., 2002), AlphaGo (Silver et al., 2016; 2017)	Exceptional AGI not yet achieved
Level 5: Superhuman <i>outperforms 100% of humans</i>	Superhuman Narrow AI AlphaFold (Jumper et al., 2021; Varadi et al., 2021), AlphaZero (Silver et al., 2018), StockFish (Stockfish, 2023)	Artificial Superintelligence (ASI) not yet achieved

What is AGI?

- Anthropic's view
 - CEO Dario Amodei prefers *powerful AI* to avoid sci-fi connotations and focuses on AI that can substantially outperform current models
 - A system that is smarter than a Nobel Prize winner across most relevant fields (biology, programming, math, writing)
 - Acts as *a country of geniuses in a data center* (analogy)
- Google's focus
 - Generalization ability: AGI can transfer knowledge and skills learned in one domain to another, enabling it to adapt to new and unseen situations effectively
 - Common sense knowledge: AGI has a vast repository of knowledge about the world, including facts, relationships, and social norms, allowing it to reason and make decisions based on this common understanding

What is Machine Learning?

- Tom Mitchell's definition
 - Improve on task T , with respect to performance metric P , based on experience E
 - T : classification, regression, detection, ...
 - P : error rate, accuracy, likelihood, margin ...
 - E : data
- Examples
 - T : Playing chess
 - P : Percentage of games won against an opponent
 - E : Playing practice games against itself

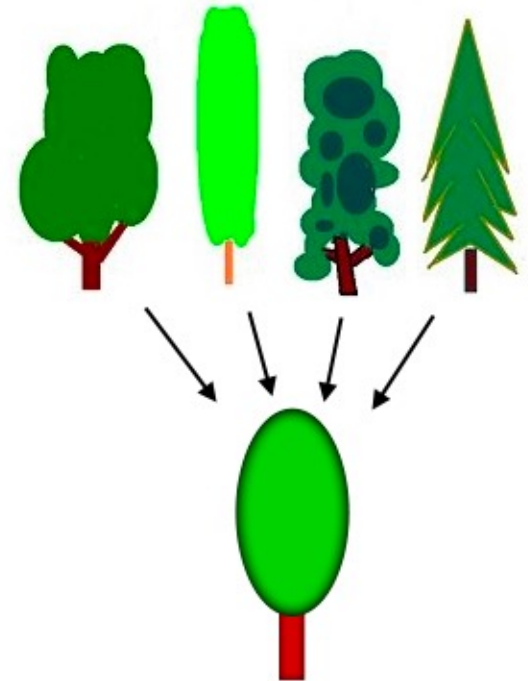


A computer program is said to learn from experience E with respect to some class of tasks T and performance measure P , if its performance at tasks in T , as measured by P , improves with experience E



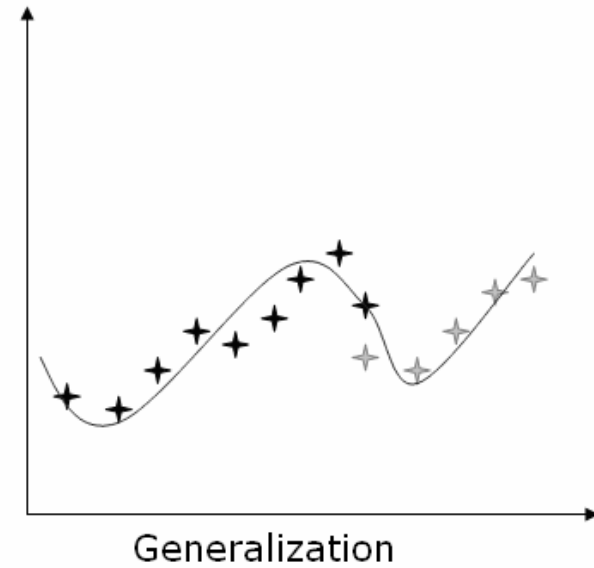
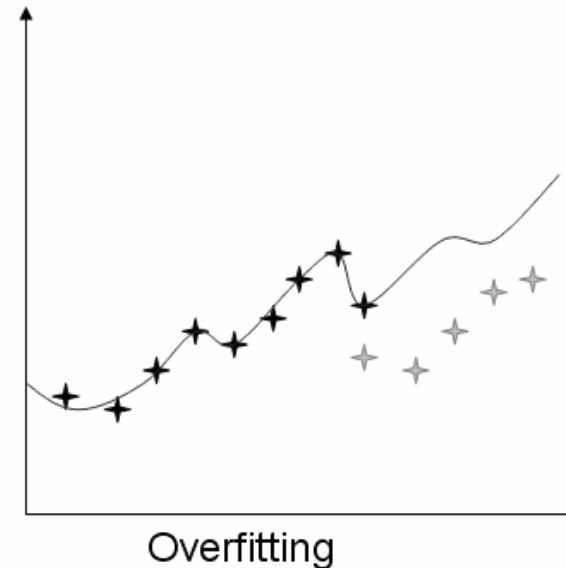
Generalization

- Objective of learning
 - Not to learn an exact representation of the training data itself
 - To build a statistical model of the process that generates the data
- Generalization
 - A form of abstraction where common properties of specific instances are formulated as general concepts or claims
 - It extracts the essence of a concept based on its analysis of similarities from many discrete objects
 - If a toddler had never seen a willow tree or pine tree before he still might classify it as a tree because it is green

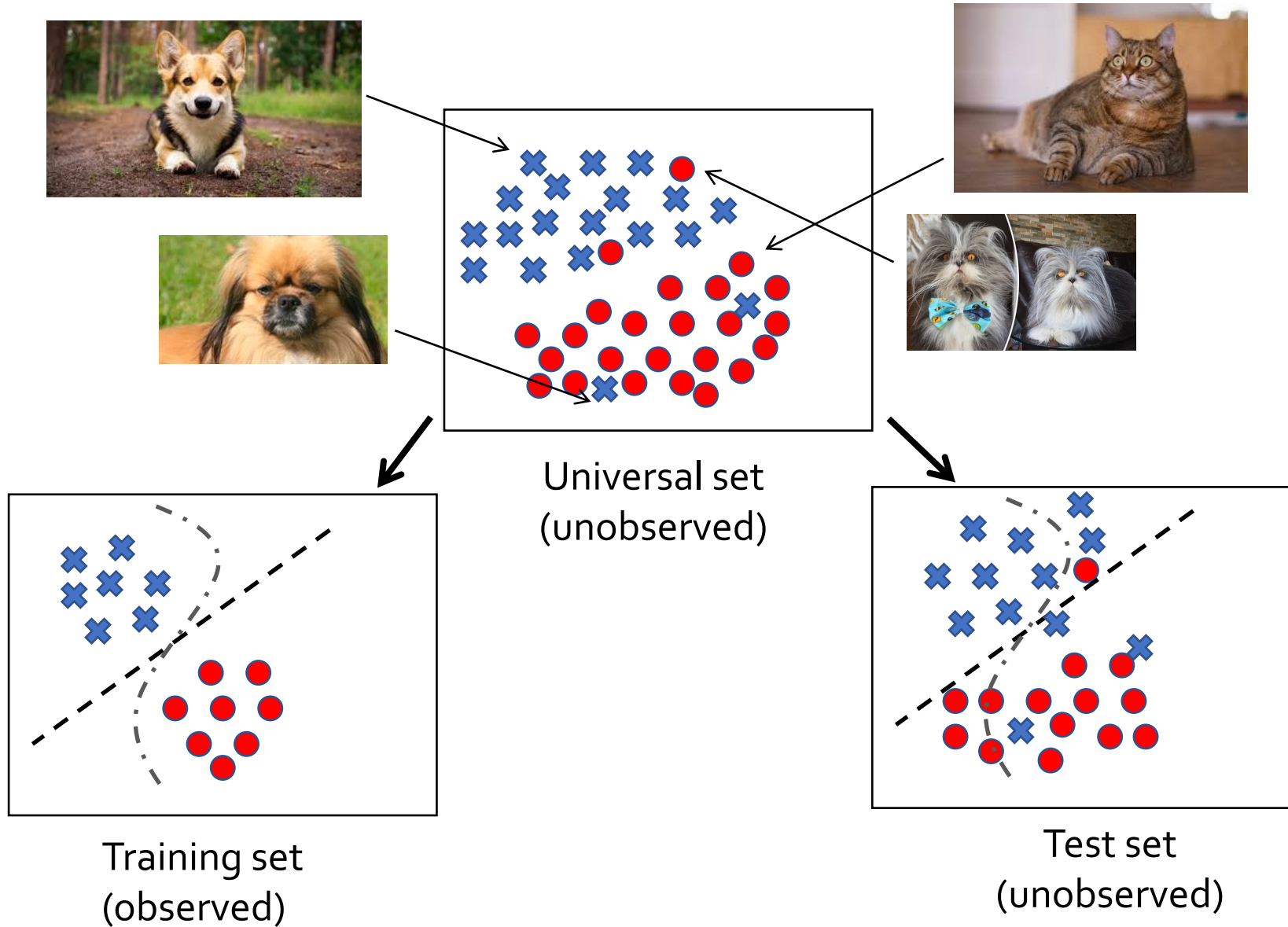


Generalization

- Generalization in ML
 - An ML model's ability to perform well on new unseen data rather than just the data that it was trained on
 - Learning algorithm maximizes accuracy on training examples
 - How to generalize from training to test?
- Strongly related to the concept of overfitting
 - Overfitting = poor generalization



Training Data vs Test Data



No Free Lunch Theorem for ML

- No machine learning algorithm is universally any better than any other
- Do not try to seek a universal learning algorithm
(No absolute best algorithm)

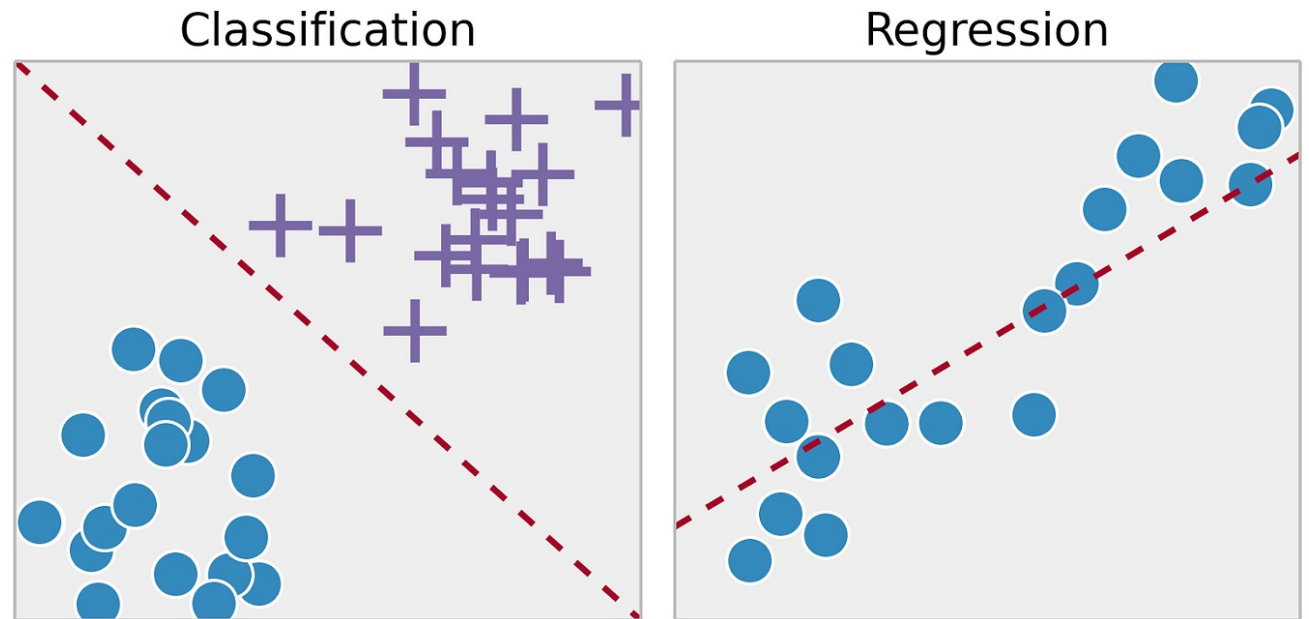


Types of Learning

- 1. Supervised learning
 - Training data includes desired outputs
- 2. Unsupervised learning
 - Training data does not include desired outputs
- 3. Semi-supervised learning
 - Some of training data includes desired outputs
- 4. Reinforcement learning
 - No fixed dataset but an environment
 - Rewards from sequence of actions

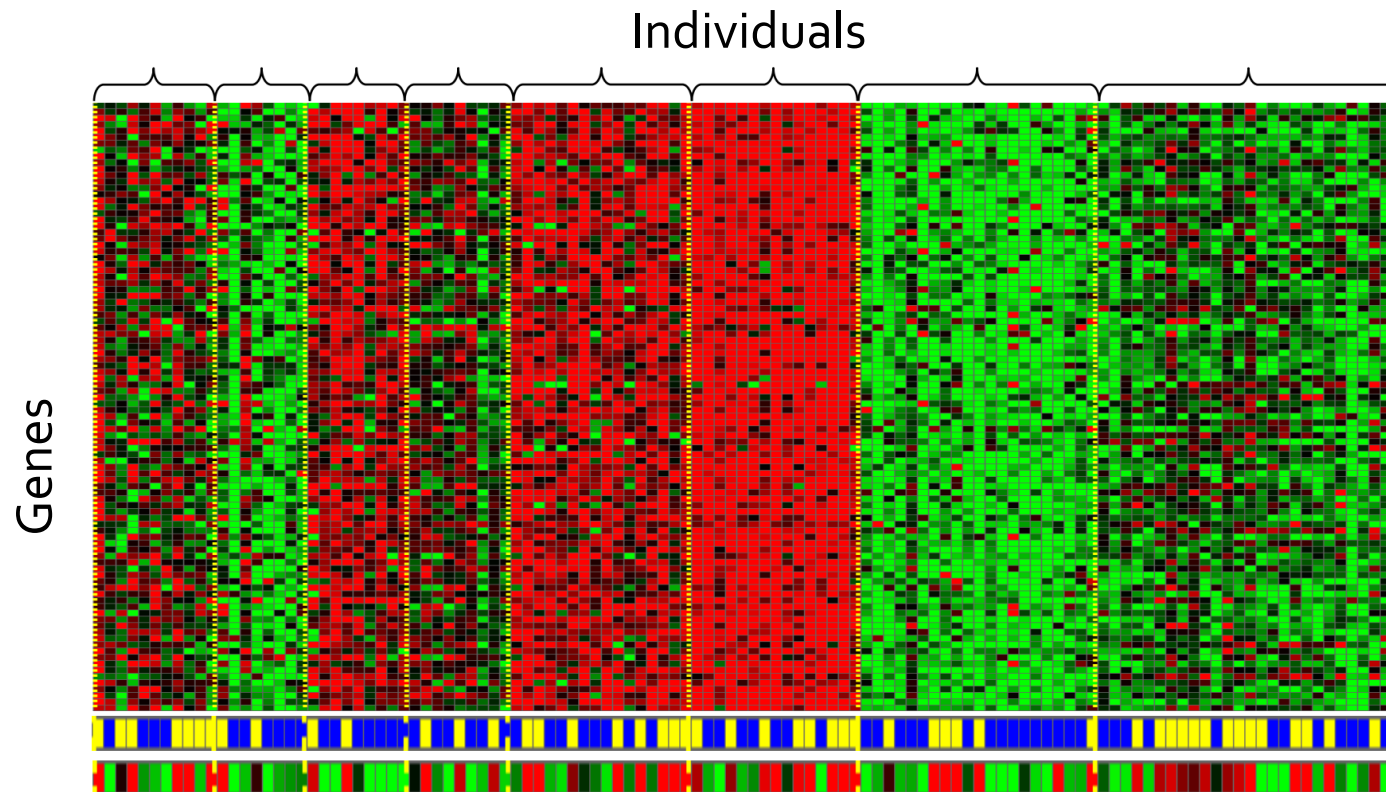
1. Supervised Learning

- Given $(\mathbf{x}_1, y_1), (\mathbf{x}_2, y_2), \dots, (\mathbf{x}_n, y_n)$, learn a function $f(\mathbf{x})$ to predict y given $\mathbf{x}$
 - Classification: y is categorical
 - Regression: y is continuous
 - $\mathbf{x}$ can be multi-dimensional (each dimension corresponds to an attribute)



2. Unsupervised Learning

- Given $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n$, find a hidden structure (e.g., clusters)
 - Clustering, anomaly detection, density estimation, ...
 - e.g., Genomics: group individuals by genetic similarity

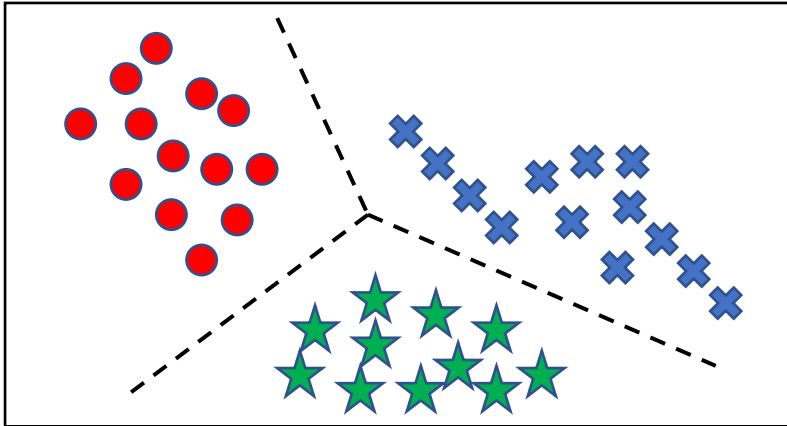


3. Semi-supervised Learning

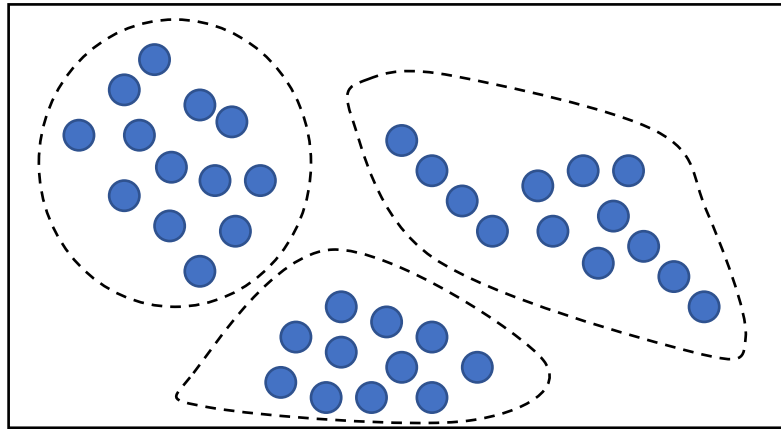
- Some of training data includes desired outputs
 - One of the bottlenecks of learning is the labeling of examples
 - Often done manually and time consuming
 - Can we label only a small number of examples and make use of a large number of unlabeled examples to learn? Possible in many cases!
- Two scenarios
 - **LU learning**: Learning with a small set of **L**abeled examples and a large set of **U**nabeled examples
 - **PU learning**: Learning with **P**ositive and **U**nabeled examples (no labeled negative examples)

Why Semi-supervised Learning Helpful?

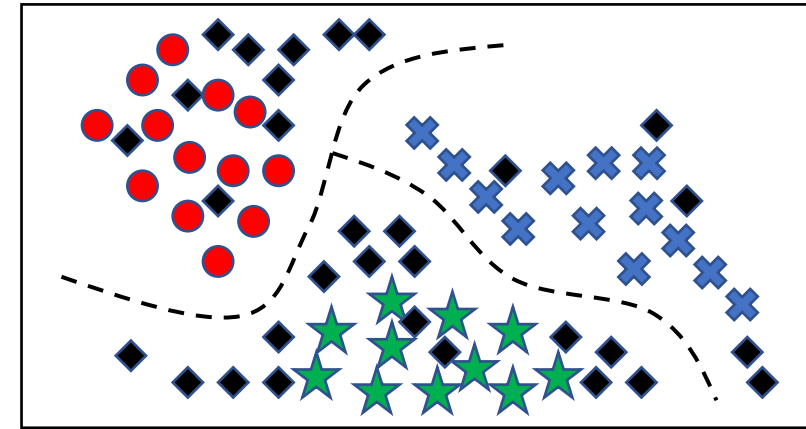
- Similar data points have similar labels
 - Unlabeled data can help identify the boundary more accurately



Supervised learning



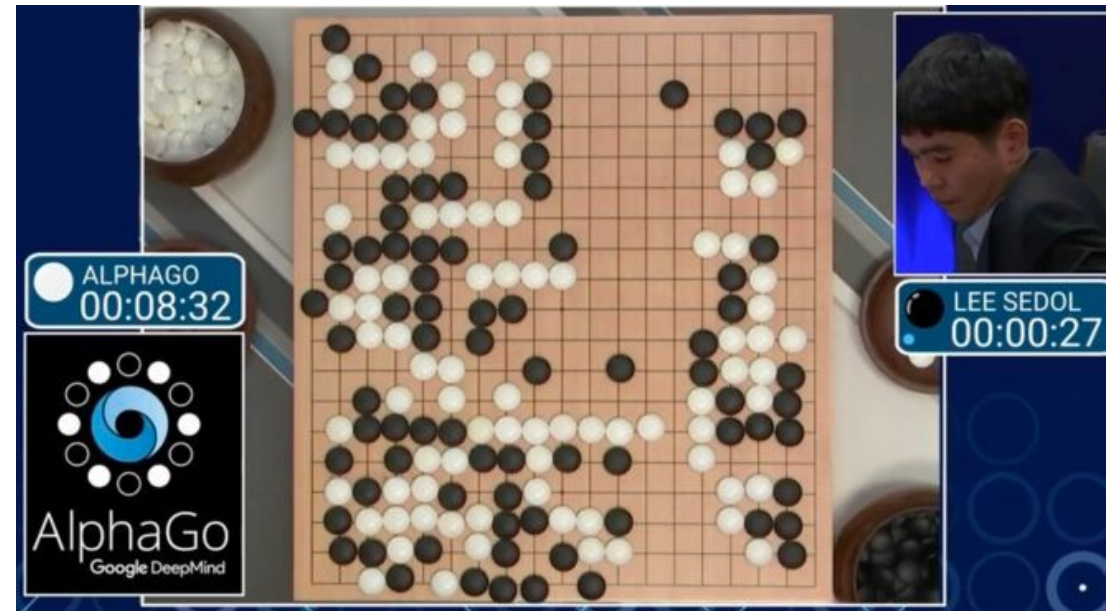
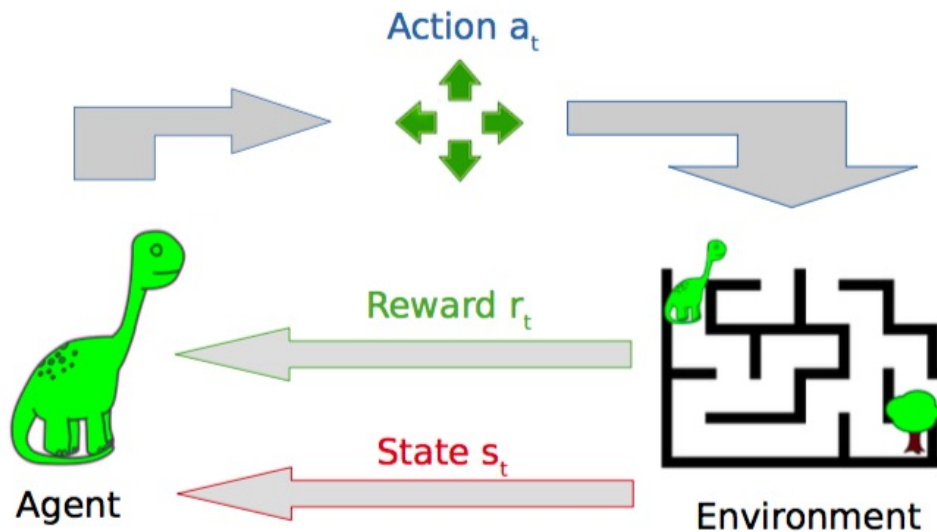
Unsupervised learning



Semi-supervised learning

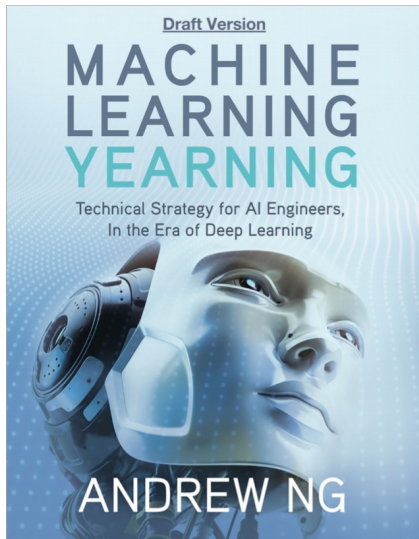
4. Reinforcement Learning

- A feedback loop between the learning system and its environment
 - Does not experience a fixed dataset
 - No supervisor, but only rewards (Learning “fills in the details”)
 - Feedback could be delayed, not instantaneous
 - Agent's actions affect the subsequent data it receives

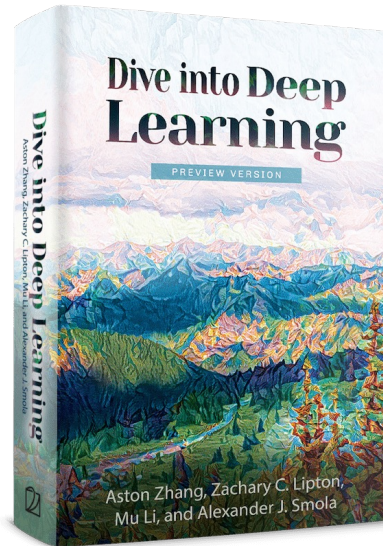


Reference for Practical ML

- Many great books, awesome tutorials, and open-source codes online



ML-Yearning, A. Ng
<https://www.deeplearning.ai/>



Dive into DL
A. Zhang et al.
(from Amazon)
<https://d2l.ai/>



Google's ML Guides (Rules of ML)
<https://developers.google.com/machine-learning/guides>

Before Machine Learning

- Don't be afraid to launch a product without machine learning
 - Machine learning is cool, but it requires large amount of data and compute
 - Need to collect data for every different problem
 - ML models would likely underperform basic heuristics
 - However, a complex heuristic is unmaintainable
- Take the time to understand your data
 - Do some exploratory data analysis (EDA)
 - Look for missing or inconsistent records; it is much easier to do this now, before you train a model, rather than later

Before Machine Learning

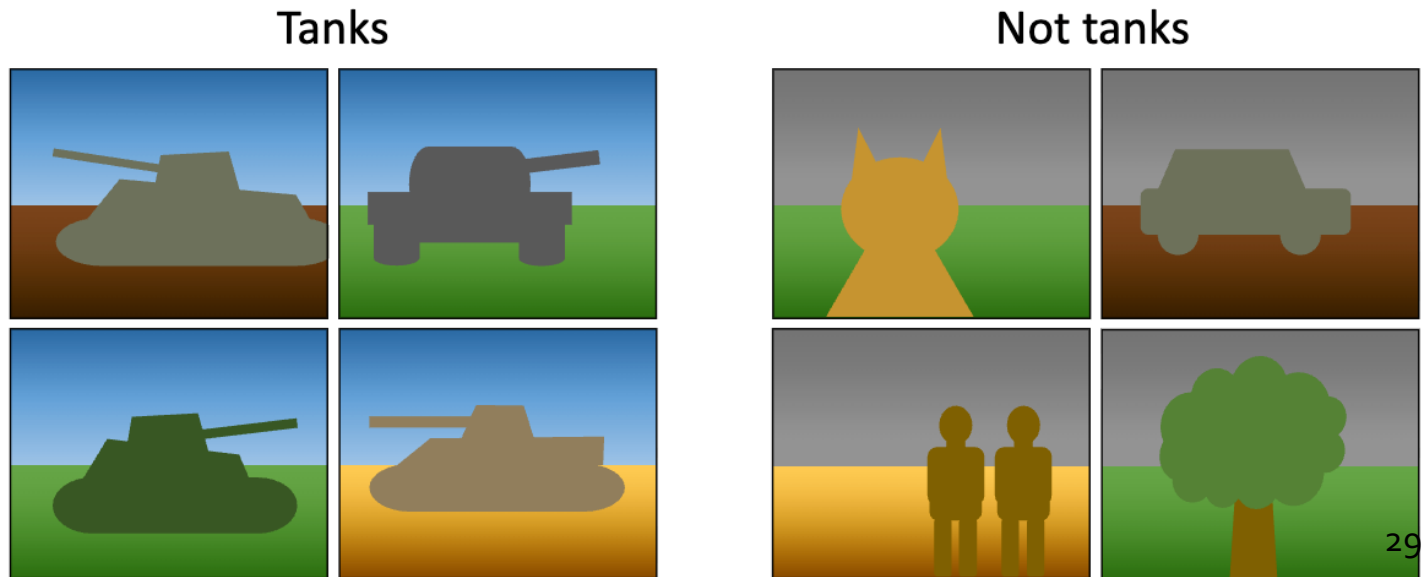
- Talk to domain experts and survey the literature
 - They can help you to understand which problems are useful to solve, and to choose the most appropriate feature set and ML model
 - You are probably not the first person to throw ML at a particular problem domain, so it is important to understand what has and has not been done previously
 - Keep up with progress in deep learning (and its pitfalls)
- First, design and implement metrics
 - Add a set of metrics to track them
 - Measure the delta between models

The First ML Pipeline

- Keep the first model simple and get the infrastructure right
 - Starting with an interpretable model makes debugging easier
 - Don't assume deep learning will be the best approach
 - No Free Lunch theorem: no ML approach is universally better than any other
 - Don't expect that the model you are working on now will be the last one
 - Start with directly observed and reported features as opposed to learned features
- Turn heuristics into features, or handle them externally
 - The problems that ML is trying to solve are not completely new
 - These same heuristics can give you a lift when tweaked with ML

The First ML Pipeline

- Don't allow test data to leak into the training process
 - When this happens, the data no longer provides a reliable measure of generality
 - This is a common reason why published ML models fail to run for real world data
- Avoid learning spurious correlations
 - E.g., Tank classification: they can be classified by merely looking at the colors of pixels in the sky, rather than recognizing the objects in the images



Iterations

- Look for patterns in the measured errors, and create new features
 - Once you have examples that the model got wrong, look for trends that are outside your current feature set
 - If you give the model a feature that allows it to fix the error, the model will try to use it
- Try to quantify observed undesirable behavior
 - Measure first, optimize second: If your issues are measurable, then you can start using them as features, objectives, or metrics

Sora 2 Is Shutting Down

- Video generation AI by OpenAI
 - Create highly realistic, consistent, and long-form video clips (and audio) from simple text or image prompts
 - Shutting down Sora web and app experiences on April 26, 2026, followed by the API on September 24, 2026
 - Expensive to run, with low monetization
 - *Focus on world simulation research to advance robotics that will help people solve real-world, physical tasks*



Sora 2 Is Shutting Down

- OpenAI is in big trouble
 - Promised adult mode - now shelved
 - Launched Sora video generator, Disney's \$1 billion equity investment in OpenAI - ended Sora 100 days later
 - Announced Stargate project - cancelled one year later
 - Altman once called AI + ads a "last resort" - 16 months later launched ads
 - Launched in-app shopping with direct checkout - now cancelled
 - Promised first hardware device this year - now delayed to 2027

Newsletters

The Atlantic

TECHNOLOGY

OpenAI Is Doing Everything ... Poorly

The company's sudden decision to pull the plug on Sora is a sign of deeper trouble.